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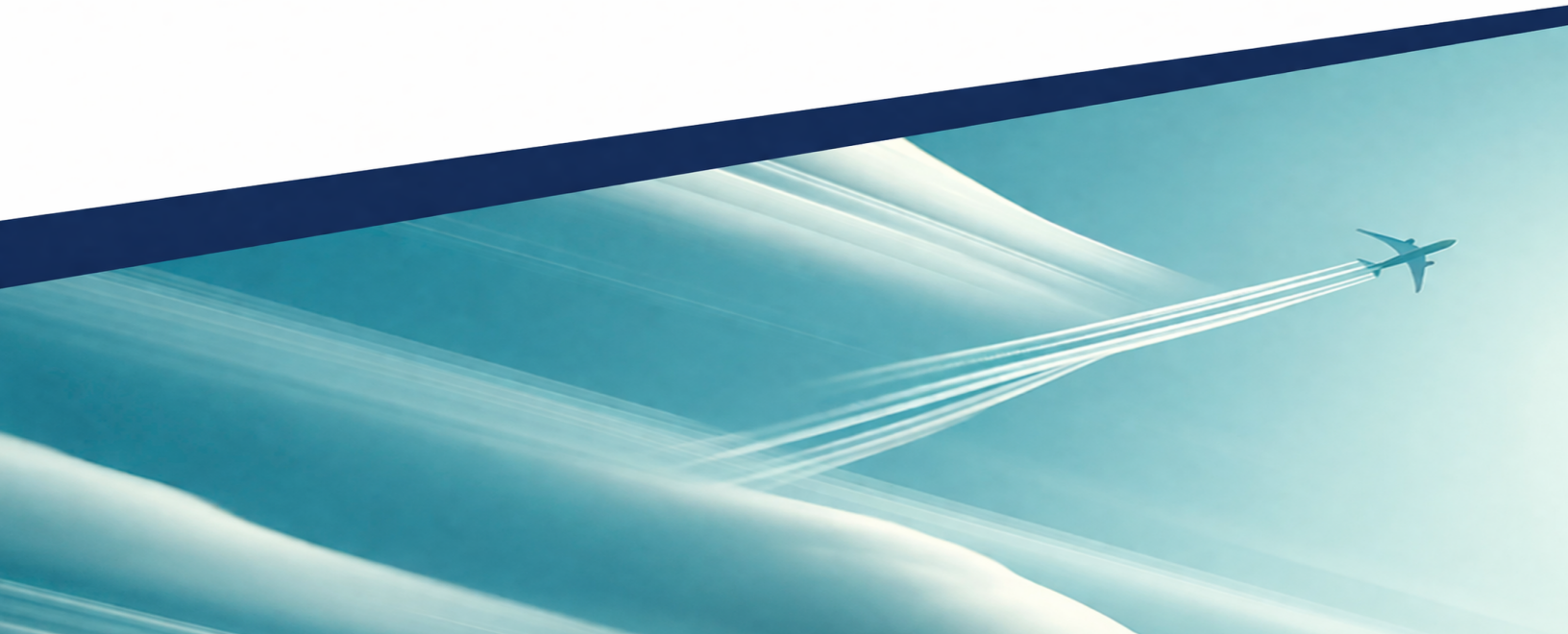
ASSIGNMENT 1

STRUCTURAL DYNAMICS AND AEROELASTICITY

Analytical Report

POLITECNICO DI MILANO
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Problem Setup

The wing has two DOFs — bending $w(y)$ and torsion $\theta(y)$ — with a strut at $y = y_B$, aerodynamic center offset e from the elastic axis, and a tip aileron of span b_a and deflection β . The Ritz–Galerkin method is applied to perform a static aeroelastic analysis.

Divergence Dynamic Pressure q_D - Derivation

Shape Functions

Root clamp imposes $w(0) = w'(0) = \theta(0) = 0$. The Ritz ansatz $w = N_w(y) \phi_w$, $\theta = N_\theta(y) \phi_\theta$ requires N_w quadratic and N_θ linear. Applying BCs and normalizing to unity at the tip :

$$\boxed{N_w(y) = \left(\frac{y}{b}\right)^2, \quad N_\theta(y) = \frac{y}{b}} \quad (1)$$

Principle of Virtual Work

In order to compute q_D , we need to find the equations of motion. To do so, we use the PVW : We need to compute the internal and external virtual work.

Internal Virtual Work

Substituting $N_w'' = 2/b^2$ and $N_\theta' = 1/b$:

$$\delta W_i = \int_0^b \delta w_{,yy} EJ w_{,yy} dy + \int_0^b \delta \theta_{,y} GJ \theta_{,y} dy = \delta \phi_w \frac{4EJ}{b^3} \phi_w + \delta \phi_\theta \frac{GJ}{b} \phi_\theta \quad (2)$$

$$\boxed{\delta W_i = \delta \phi^T \begin{bmatrix} K_w & 0 \\ 0 & K_\theta \end{bmatrix} \phi, \quad K_w = \frac{4EJ}{b^3}, \quad K_\theta = \frac{GJ}{b}} \quad (3)$$

External Virtual Work

$$\delta W_e = \delta W_{\text{AERO}}^{\text{wing}} + \delta W_{\text{AERO}}^{\text{aileron}} + \delta W_{\text{STRUT}} + \delta W_{\text{INERTIA}} + \delta W_{\text{GRAVITY}}$$

Since we are interested in the divergence dynamic pressure q_D , we can neglect inertia and gravity (static case, no external forces). Thus :

$$\boxed{\delta W_e = \delta W_A^w + \delta W_A^a + \delta W_{\text{STR}}}$$

(1) Wing aerodynamic virtual work δW_A^w The kinematically admissible AC displacement is $h_{AC} = w + e\theta$, and strip lift reads $L'_w = q c C_{L/\alpha} \theta$ (perturbation, $\alpha_{\text{eff}} = \theta$, $C_{L_0} = 0$). Thus :

$$\delta W_A^w = \int_0^b \delta h_{AC} L'_w dy = \int_0^b \delta \phi_w \left(\frac{y}{b}\right)^2 q c C_{L/\alpha} \frac{y}{b} \phi_\theta dy + \int_0^b \delta \phi_\theta \frac{y}{b} e q c C_{L/\alpha} \frac{y}{b} \phi_\theta dy \quad (4)$$

$$\boxed{\delta W_A^w = \delta \phi^T q \mathbf{K}_A \phi}, \quad \mathbf{K}_A = \begin{bmatrix} 0 & \frac{cb}{4} C_{L/\alpha} \\ 0 & \frac{ecb}{3} C_{L/\alpha} \end{bmatrix} \quad (5)$$

$\mathbf{K}_A$ is the aerodynamic stiffness matrix.

(2) Aileron aerodynamic virtual work δW_A^a Not needed for q_D (homogeneous problem) but derived for later use. The aileron spans $[b - b_a, b]$ with deflection β ; perturbation gives $C_{m_{AC}} = C_{m/\beta} \beta$. The virtual work splits into a lift contribution and a pitching moment contribution :

$$\delta W_A^a = \int_{b-b_a}^b \delta(w + e\theta) q c C_{L/\beta} \beta dy + \int_{b-b_a}^b \delta\theta q c^2 C_{m/\beta} \beta dy \quad (6)$$

$$\boxed{\delta W_A^a = \delta \phi^T q \beta \mathbf{F}_a}, \quad \mathbf{F}_a = \frac{c}{b} \left\{ \begin{array}{c} C_{L/\beta} \cdot \frac{b^3 - (b - b_a)^3}{3b} \\ (e C_{L/\beta} + c C_{m/\beta}) \frac{b^2 - (b - b_a)^2}{2} \end{array} \right\} = \begin{Bmatrix} F_{a1} \\ F_{a2} \end{Bmatrix} \quad (7)$$

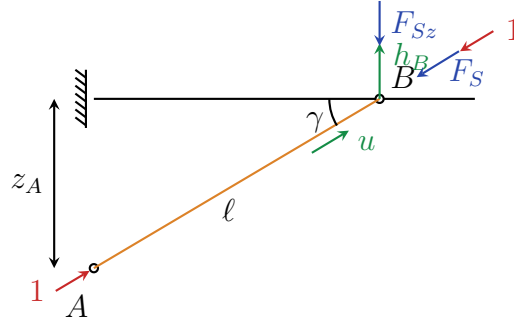


FIGURE 1 – Equivalent model of the strut, replaced by an equivalent axial spring of stiffness K_{STR} .

(3) Strut virtual work δW_{STR} The strut axial force and its z -component follow from geometry ($\ell = z_A / \sin \gamma$, $u = h_B \sin \gamma$) :

$$\boxed{F_{Sz} = - \underbrace{\frac{EA}{z_A}}_{K_{STR}} \sin^3 \gamma h_B} \quad (8)$$

The displacement at attachment point B, using $\eta_B = z_A / (b \tan \gamma)$:

$$h_B = w(y_B) - x_B \theta(y_B) = \eta_B [\eta_B \phi_w - x_B \phi_\theta]$$

Substituting into $\delta W_{STR} = \delta h_B (-K_{STR}) h_B$:

$$\boxed{\delta W_{STR} = -\delta \phi^T K_{STR} \eta_B^2 \begin{bmatrix} \eta_B^2 & -\eta_B x_B \\ -\eta_B x_B & x_B^2 \end{bmatrix} \phi} \quad (9)$$

Application of PVW : $\delta W_i = \delta W_e$

$$\delta \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix}^T \begin{bmatrix} K_w & 0 \\ 0 & K_\theta \end{bmatrix} \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} = \delta \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix}^T \left(q \mathbf{K}_A \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} + q \beta \mathbf{F}_a - K_{\text{STR}} \eta_B^2 \begin{bmatrix} \eta_B^2 & -\eta_B x_B \\ -\eta_B x_B & x_B^2 \end{bmatrix} \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} \right) \quad (10)$$

Assembling all contributions and invoking the arbitrariness of $\delta \phi$ (unforced case $\beta = 0$ for q_D computation) :

$$\left(\begin{bmatrix} K_w + K_{\text{STR}} \eta_B^4 & -K_{\text{STR}} \eta_B^3 x_B \\ -K_{\text{STR}} \eta_B^3 x_B & K_\theta + K_{\text{STR}} \eta_B^2 x_B^2 \end{bmatrix} - q \mathbf{K}_A \right) \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \quad (11)$$

This is $(\mathbf{K}_S - q_D \mathbf{K}_A) \phi = \mathbf{0}$, with :

$$\mathbf{K}_S = \begin{bmatrix} K_{S_1} & K_{S_{12}} \\ K_{S_{12}} & K_{S_2} \end{bmatrix}, \quad \mathbf{K}_A = \begin{bmatrix} 0 & \mathbf{K}_{A_1} = \frac{cb}{4} C_{L/\alpha} \\ 0 & \mathbf{K}_{A_2} = \frac{ecb}{3} C_{L/\alpha} \end{bmatrix} \quad (12)$$

$$K_{S_1} = \frac{4EJ}{b^3} + \frac{EA z_A^3}{b^4} \frac{\cos^4 \gamma}{\sin \gamma}, \quad K_{S_{12}} = -\frac{EA z_A^2}{b^3} \cos^3 \gamma x_B, \quad K_{S_2} = \frac{GJ}{b} + \frac{EA \sin \gamma \cos^2 \gamma z_A x_B^2}{b^2} \quad (13)$$

For divergence, the condition is : $\det(\mathbf{K}_S - q_D \mathbf{K}_A) = 0$. Expanding the determinant :

$$\det(\mathbf{K}_S - q_D \mathbf{K}_A) = K_{S_1} K_{S_2} - q_D K_{S_1} K_{A_2} - K_{S_{12}}^2 + q_D K_{S_{12}} K_{A_1} = 0 \quad \Rightarrow \quad q_D = \frac{K_{S_1} K_{S_2} - (K_{S_{12}})^2}{K_{S_1} K_{A_2} - K_{S_{12}} K_{A_1}} \quad (14)$$

Control Effectiveness Computation

The control effectiveness is defined as $E_C = \frac{L}{L_R}$. The total lift is obtained by spanwise integration :

$$L = \int_0^b L'(y) dy = qc \left(\int_0^b C_{L/\alpha} \underbrace{\theta(y)}_{\frac{y}{b} \phi_\theta} dy + \int_{b-b_a}^b C_{L/\beta} \beta dy \right)$$

$$L = \underbrace{qcb_a C_{L/\beta} \beta}_{L_R \text{ (Rigid wing)}} + \underbrace{qc \frac{b C_{L/\alpha}}{2} \phi_\theta}_{L_E \text{ (Elastic torsion)}} \quad (15)$$

We need to determine ϕ_θ . From the first question, we have :

$$(\mathbf{K}_S - q \mathbf{K}_A) \begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} = q \beta \mathbf{F}_a \quad \text{where we define } \mathbf{K}_{\text{TOT}} = \mathbf{K}_S - q \mathbf{K}_A \quad (16)$$

For $q < q_D$, $\mathbf{K}_{\text{TOT}}$ is invertible :

$$\mathbf{K}_{\text{TOT}}^{-1} = \frac{1}{\det(\mathbf{K}_{\text{TOT}})} \begin{bmatrix} K_{\text{TOT},22} & -K_{\text{TOT},12} \\ -K_{\text{TOT},21} & K_{\text{TOT},11} \end{bmatrix} \quad (17)$$

Where $\det(\mathbf{K}_{\text{TOT}})$ is defined at equation (14). Thus :

$$\begin{Bmatrix} \phi_w \\ \phi_\theta \end{Bmatrix} = q\beta \mathbf{K}_{\text{TOT}}^{-1} \mathbf{F}_a \quad (18)$$

The aileron influence load vector $\mathbf{F}_a$ is defined at (7). From matrix multiplication :

$$\phi_\theta = \frac{q\beta}{\det(\mathbf{K}_{\text{TOT}})} (K_{\text{TOT},11} F_{a2} - K_{\text{TOT},21} F_{a1}) \quad (19)$$

Substituting into the lift expression :

$$L = qc\beta \left(b_a C_{L/\beta} + \frac{b C_{L/\alpha}}{2} \cdot \frac{q}{\det(\mathbf{K}_{\text{TOT}})} (K_{\text{TOT},11} F_{a2} - K_{\text{TOT},21} F_{a1}) \right) \quad (20)$$

Dividing by L_R gives the control effectiveness E_C :

$$E_C = 1 + q \cdot \frac{b}{b_a} \cdot \frac{C_{L/\alpha}}{C_{L/\beta}} \cdot \frac{1}{2 \det(\mathbf{K}_{\text{TOT}})} \cdot (K_{\text{TOT},11} F_{a2} - K_{\text{TOT},21} F_{a1}) \quad (21)$$

The expression of E_C separates the rigid contribution from the aeroelastic correction. At $q = 0$, one recovers $E_C = 1$ (rigid limit); as q increases, aileron-induced torsion modifies total lift. The correction is amplified as $\det(\mathbf{K}_{\text{TOT}}) \rightarrow 0$ (divergence), so E_C may vanish before divergence – defining the control reversal pressure.

Wing deformed shape – Strut effect

Compare wing deformation for $\beta_0 = 10^\circ$ at $q = 0.4 q_D^{(0)}$, where $q_D^{(0)}$ is the divergence dynamic pressure without strut.

Displacement field.

$$z(x, y) = w(y) - x \theta(y) = \left(\frac{y}{b} \right)^2 \phi_w - x \frac{y}{b} \phi_\theta \quad (22)$$

Aeroelastic equilibrium. Inverting $\mathbf{K}_{\text{TOT}} \{\phi_w, \phi_\theta\}^\top = q\beta \mathbf{F}_a$ gives

$$\phi_w = \frac{q\beta}{\det \mathbf{K}_{\text{TOT}}} (K_{\text{TOT},22} F_{a1} - K_{\text{TOT},12} F_{a2}), \quad (23)$$

$$\phi_\theta = \frac{q\beta}{\det \mathbf{K}_{\text{TOT}}} (K_{\text{TOT},11} F_{a2} - K_{\text{TOT},21} F_{a1}), \quad (24)$$

hence

$$z(x, y) = \frac{q\beta}{\det \mathbf{K}_{\text{TOT}}} \frac{y}{b} \left[\frac{y}{b} (K_{\text{TOT},22} F_{a1} - K_{\text{TOT},12} F_{a2}) - x (K_{\text{TOT},11} F_{a2} - K_{\text{TOT},21} F_{a1}) \right]. \quad (25)$$

Configurations compared.

- **With strut** : full structural model ;
- **Without strut** : strut axial stiffness set to $EA = 0$.

To achieve the comparison we define :

$$\delta_z = 100 \frac{|z_{\text{strut}} - z_{\text{ns}}|}{|z_{\text{ns}}|}. \quad (26)$$

Which is the relative deviation of the strut configuration with respect to the no-strut reference.

Convergence of q_D w.r.t. Ritz shape functions number

To study convergence of the Ritz approximation, bending and torsional displacements are expanded on a hierarchical polynomial basis that satisfies the clamped-root essentials conditions $w(0) = w'(0) = \theta(0) = 0$:

$$w(y) = \sum_{i=1}^N \left(\frac{y}{b}\right)^{i+1} \phi_{w_i} = \underline{\mathbf{N}}_w^T \underline{\phi}_w, \quad \underline{\mathbf{N}}_w = \{\eta^{i+1}\}_N, \quad \underline{\phi}_w = \{\phi_{w_i}\}_N, \quad \eta = \frac{y}{b} \quad (27)$$

$$\theta(y) = \sum_{i=1}^N \left(\frac{y}{b}\right)^i \phi_{\theta_i} = \underline{\mathbf{N}}_\theta^T \underline{\phi}_\theta, \quad \underline{\mathbf{N}}_\theta = \{\eta^i\}_N, \quad \underline{\phi}_\theta = \{\phi_{\theta_i}\}_N. \quad (28)$$

Internal virtual work

Substituting the Ritz ansatz into $\delta W_i = \int_0^b \delta w'' EJ w'' dy + \int_0^b \delta \theta' GJ \theta' dy$ and factoring out the generalized DOFs :

$$\delta W_i = \delta \underline{\Phi}^T \underline{\mathbf{K}}_{\text{WSTIF}}^{(N)} \underline{\Phi}, \quad \underline{\mathbf{K}}_{\text{WSTIF}}^{(N)} = \begin{bmatrix} \underline{\mathbf{K}}_w^{(N)} & \underline{\mathbf{0}} \\ \underline{\mathbf{0}} & \underline{\mathbf{K}}_\theta^{(N)} \end{bmatrix}, \quad \underline{\Phi} = \left\{ \begin{matrix} \underline{\phi}_w \\ \underline{\phi}_\theta \end{matrix} \right\}, \quad (29)$$

where the stiffness sub-matrices are

$$\underline{\mathbf{K}}_w^{(N)} = \int_0^b \underline{\mathbf{N}}_w'' EJ \underline{\mathbf{N}}_w''^T dy, \quad \underline{\mathbf{K}}_\theta^{(N)} = \int_0^b \underline{\mathbf{N}}_\theta' GJ \underline{\mathbf{N}}_\theta'^T dy, \quad (30)$$

with

$$\underline{\mathbf{N}}_w'' = \frac{1}{b^2} \{(i+1)i\eta^{i-1}\}_N, \quad \underline{\mathbf{N}}_\theta' = \frac{1}{b} \{i\eta^{i-1}\}_N. \quad (31)$$

Aerodynamic virtual work

The AC displacement is $h_{AC} = \underline{\mathbf{N}}_w^T \underline{\phi}_w + e \underline{\mathbf{N}}_\theta^T \underline{\phi}_\theta$ and strip lift reads $L' = qcC_{L/\alpha} \underline{\mathbf{N}}_\theta^T \underline{\phi}_\theta$. Expanding $\delta W_{e,A}^w = \int_0^b \delta h_{AC} L' dy$ gives :

$$\delta W_A^w = \delta \underline{\Phi}^T q \underline{\mathbf{K}}_A^{(N)} \underline{\Phi}, \quad \underline{\mathbf{K}}_A^{(N)} = \begin{bmatrix} \underline{\mathbf{0}} & \underline{\mathbf{K}}_{A1}^{(N)} \\ \underline{\mathbf{0}} & \underline{\mathbf{K}}_{A2}^{(N)} \end{bmatrix}, \quad (32)$$

$$\underline{\mathbf{K}}_{A1}^{(N)} = \int_0^b \underline{\mathbf{N}}_w cC_{L/\alpha} \underline{\mathbf{N}}_\theta^T dy, \quad \underline{\mathbf{K}}_{A2}^{(N)} = \int_0^b \underline{\mathbf{N}}_\theta e cC_{L/\alpha} \underline{\mathbf{N}}_\theta^T dy. \quad (33)$$

Strut contribution

The strut stiffness $K_{\text{STR}} = (EA/z_A) \sin^3 \gamma$ is unchanged. The attachment-point displacement

$$h_B = \underline{\mathbf{N}}_w^T(\eta_B) \underline{\phi}_w - x_B \underline{\mathbf{N}}_\theta^T(\eta_B) \underline{\phi}_\theta, \quad \eta_B = \frac{z_A}{b \tan \gamma}, \quad (34)$$

enters $\delta W_{\text{STR}} = -\delta h_B K_{\text{STR}} h_B$. Substituting the Ritz approximation and expanding :

$$\boxed{\delta W_{\text{STR}} = -\delta \underline{\Phi}^T \underline{\mathbf{K}}_{\text{STR}}^{(N)} \underline{\Phi}}, \quad (35)$$

$$\underline{\mathbf{K}}_{\text{STR}}^{(N)} = \begin{bmatrix} \underline{\mathbf{K}}_{\text{STR},11}^{(N)} & \underline{\mathbf{K}}_{\text{STR},12}^{(N)} \\ \underline{\mathbf{K}}_{\text{STR},21}^{(N)} & \underline{\mathbf{K}}_{\text{STR},22}^{(N)} \end{bmatrix} = \begin{bmatrix} \underline{\mathbf{N}}_w(\eta_B) K_{\text{STR}} \underline{\mathbf{N}}_w^T(\eta_B) & -x_B K_{\text{STR}} \underline{\mathbf{N}}_w(\eta_B) \underline{\mathbf{N}}_\theta^T(\eta_B) \\ -x_B K_{\text{STR}} \underline{\mathbf{N}}_\theta(\eta_B) \underline{\mathbf{N}}_w^T(\eta_B) & x_B^2 K_{\text{STR}} \underline{\mathbf{N}}_\theta(\eta_B) \underline{\mathbf{N}}_\theta^T(\eta_B) \end{bmatrix} \quad (36)$$

Principle of virtual work and convergence criterion

Assembling all contributions, $\underline{\mathbf{K}}_S^{(N)} = \underline{\mathbf{K}}_{\text{WSTIF}}^{(N)} + \underline{\mathbf{K}}_{\text{STR}}^{(N)}$, the aeroelastic eigenproblem reads

$$\left[\underline{\mathbf{K}}_S^{(N)} - q \underline{\mathbf{K}}_A^{(N)} \right] \underline{\Phi} = \underline{\mathbf{0}}. \quad (37)$$

The divergence pressure and convergence indicator are

$$\boxed{q_D^{(N)} = \min \left\{ q > 0 \mid \det(\underline{\mathbf{K}}_S^{(N)} - q \underline{\mathbf{K}}_A^{(N)}) = 0 \right\}}, \quad \varepsilon_N = \left| \frac{q_D^{(N)} - q_D^{(N-1)}}{q_D^{(N-1)}} \right|. \quad (38)$$

Rapid stabilization of $q_D^{(N)}$ with N validates the polynomial Ritz basis.

Iterative redesign strategy for increasing the divergence pressure

Objective

Find a configuration $\mathbf{X} = (EJ, GJ, EA, \gamma)$ minimizing a mass index $\mathcal{M}(\mathbf{X})$ subject to the requirement

$$q_D(\mathbf{X}) \geq q_D^{\text{tar}} = 1.4 q_D(\mathbf{X}^{(0)}).$$

Aerodynamic parameters, x_B , and z_A are fixed. The target is computed once from the baseline $\mathbf{X}^{(0)} = (EJ_0, GJ_0, EA_0, \gamma_0)$ and remains unchanged throughout.

Mass index

Since no density data is available, a normalized relative mass indicator is introduced with unit density. The strut length is $\ell(\gamma) = z_A / \sin \gamma$, so its mass scales as $EA \ell(\gamma)$. The global index is

$$\mathcal{M}(\mathbf{X}) = \underbrace{\left(\frac{EJ}{EJ_0} - 1 \right) + \left(\frac{GJ}{GJ_0} - 1 \right)}_{\text{wing bending + torsion}} + \underbrace{\frac{EA}{EA_0} \frac{\sin \gamma_0}{\sin \gamma} - 1}_{\text{strut}},$$

with $\mathcal{M}(\mathbf{X}^{(0)}) = 0$ by construction. Lighter admissible redesigns correspond to smaller $\mathcal{M}$.

Search strategy

Let $\mathcal{P} = \{EJ, GJ, EA, \gamma\}$. At each iteration k , all 15 (4 singletons, 6 doublets, 4 triplets, 1 quadruplet) non-empty subsets $S \subset \mathcal{P}$ are explored :

- variables *in* S are varied on a discrete local grid around $\mathbf{X}^{(k)}$, with both positive and negative variations allowed within prescribed bounds ; after a feasible design is found, a parameter can be reduced if another is increased more efficiently, enabling the total mass to decrease from one iteration to the next.
- variables *not in* S are kept fixed at their current value.

Feasible set and update rule

For each subset S , the feasible candidates are

$$\mathbf{Cand}_S^{(k)} = \left\{ \mathbf{X} \in \mathcal{N}_S^{(k)} \mid q_D(\mathbf{X}) \geq q_D^{\text{tar}} \right\},$$

and the lightest feasible candidate per subset is $\mathbf{Cand}_{S,\min}^{(k)} = \min_{\mathbf{X} \in \mathbf{Cand}_S^{(k)}} \mathcal{M}(\mathbf{Cand}_S^{(k)})$. The global best of the iteration is

$$\mathbf{Cand}_{\text{best}}^{(k)} = \min_S \mathcal{M}(\mathbf{Cand}_{S,\min}^{(k)}).$$

The design is updated only if a strictly lighter feasible candidate is found :

$$\mathbf{X}^{(k+1)} = \mathbf{Cand}_{\text{best}}^{(k)} \quad \text{if} \quad \mathcal{M}(\mathbf{Cand}_{\text{best}}^{(k)}) < \mathcal{M}(\mathbf{X}^{(k)}).$$

Step refinement

If no lighter feasible candidate is found at the current grid resolution, the search steps are reduced and the procedure is repeated around the same design. Large initial steps allow rapid identification of a feasible region ; smaller steps then refine toward a local minimum of $\mathcal{M}$.

Stopping criterion

The procedure stops when both conditions hold :

1. no lighter feasible candidate is found in any of the 15 subset searches ;
2. all search steps have fallen below prescribed minimum values.

The converged solution $\mathbf{X}^* = (EJ^*, GJ^*, EA^*, \gamma^*)$ satisfies $q_D(\mathbf{X}^*) \geq q_D^{\text{tar}}$ and is the lightest feasible design found by the procedure.

Conclusions

This strategy is intentionally crude (no analytical optimization, nor use of optimization laws/principles/-theorems such as the KKT conditions) and purely numerical in order to exploit a maximum MATLAB.

The result does not guarantee a global optimum ; however, the convergence of the algorithm makes it possible to identify a configuration that minimizes the mass index while satisfying the constraint on q_D .

It should nevertheless be emphasized that the result is not exact but rather provides an order of magnitude, due to the simplifications made in the density model and in the definition of the mass index. A different definition of the mass index would lead to a different result, although still of the same order of magnitude.